

WASP-50b

R-0665

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_251229 · created 2026-07-27T18:04:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461038.8 +/- 0.14 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.049
Transit depth (Rp/Rs)^2	2.75 +/- 1.63 [%]
Orbital Inclination (inc)	83.11 +/- 2.88
Airmass coefficient 1 (a1)	0.985 +/- 0.028
Airmass coefficient 2 (a2)	0.007 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	2.81 %
Best Comparison Star	#1 - [283, 256]
Optimal Aperture	2.75
Optimal Annulus	7.82
Transit Duration (day)	0.02 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.049 vs 0.1404 (deviation 0.5σ)
Duration fit vs published	0.02 d vs 0.0754 d (deviation 1.85σ)
Mid-time O–C	89.2 min
Depth SNR	3.3
Residual scatter	2.81 %

Light curve

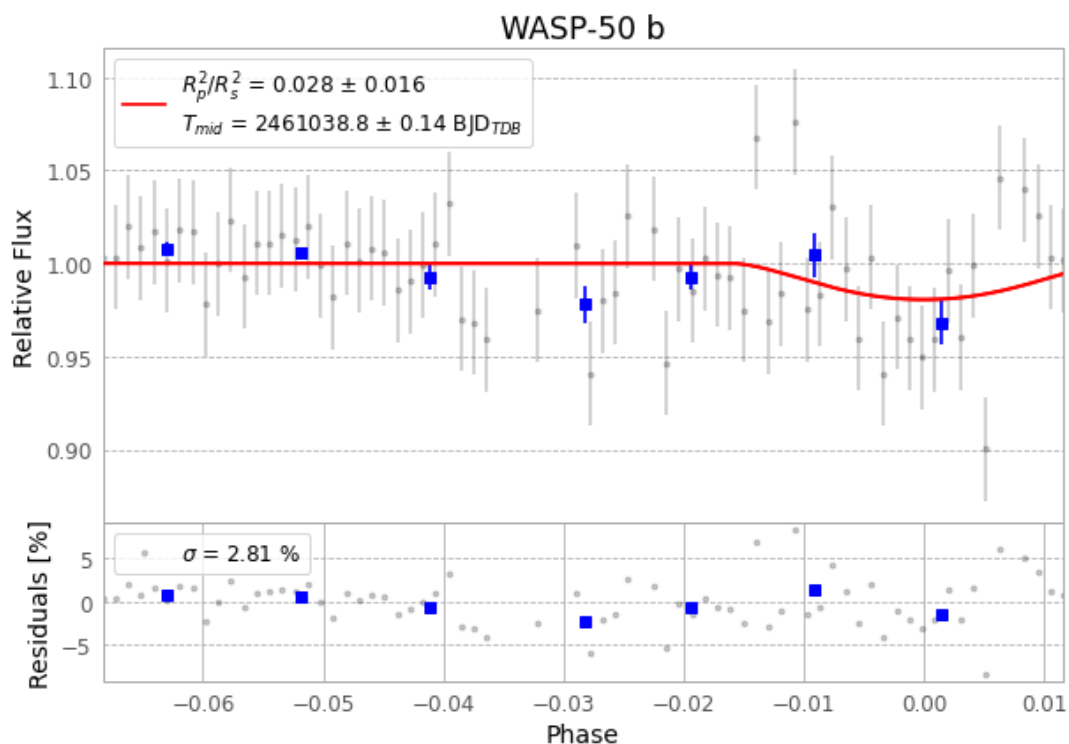


Figure 1 — FinalLightCurve_WASP-50 b_2025-12-28.png (SHA-256 8a2f44149e49c24d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 319], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 355981ff8295488b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d476b1b

Data provenance

76 FITS frames (50 MB), WASP-50251229034816.FITS → WASP-50251229073315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-251229.log (14ab975585a3...), FinalLightCurve_WASP-50 b_2025-12-28.png (8a2f44149e49...), FinalLightCurve_WASP-50 b_2025-12-28.csv (ed0f9bc1a985...)

Compute resources

Wall time 700.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 18:04Z.